

SECTION A

1 (short)

- (a) The open-ended U-tube shown in Fig. 1 is used to measure the density ρ_1 of a liquid of constant density using another liquid of known constant density ρ_2 . Both ρ_1 and ρ_2 are much larger than the density of air. Stating your reasoning, find ρ_1 in terms of ρ_2 , h_1 and h_2 . [4]
- (b) The density of a fluid (salt stratified water) is given by $\rho = 1100 + 500y \text{ kg m}^{-3}$ where y is measured in metres vertically downwards from the free surface of the fluid.
- (i) Find an expression for the gauge pressure in the fluid at depth y . [3]
- (ii) Calculate the horizontal component of the hydrostatic force per unit width acting on one side of a rectangular gate submerged in the fluid. The top of the gate is at a depth $y = 1.0 \text{ m}$ and the bottom is at a depth of $y = 2.0 \text{ m}$. [3]

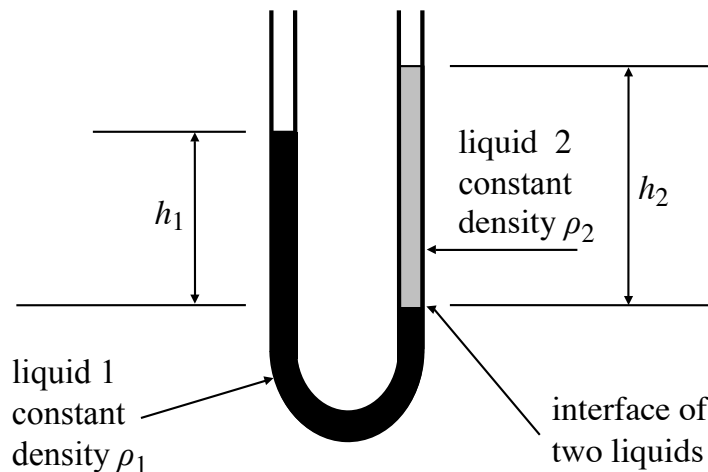


Fig. 1